

Comprehensive Molecular Electronic Structure Report via A-CASE:

End-to-End Simulation, Pipeline Architecture, and Full Chemical
Accuracy Benchmarks for LiH, BeH₂, HF, and H₂O

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Abstract

This report presents the complete theoretical architecture, implementation methodology, and empirical benchmark results for four target molecules—**Lithium Hydride (LiH)**, **Beryllium Hydride (BeH₂)**, **Hydrogen Fluoride (HF)**, and **Water (H₂O)**—simulated via the `clifford.qc` framework operating on the complexified Clifford algebra $\text{Cl}(2n, \mathbb{C}) \cong \text{M}(2^n, \mathbb{C})$. Upstream molecular integrals generated by PySCF in an STO-3G minimal basis are exported as restricted FCIDUMP records and ingested into `clifford.qc`. Ground-state energies, correlation observables ($\langle d \rangle$, $\langle S^2 \rangle$), static magnetic susceptibilities (χ_0), and dynamic Lehmann response spectra are evaluated using both the **Adaptive Clifford-Algebra Subspace Eigensolver (A-CASE)** and the **Complete Single-and-Double (SD) Coordinate Subspace Eigensolver**. **Sub-millihartree chemical accuracy** (≤ 1.5936 mHa) is achieved across all four molecules relative to exact sector Full Configuration Interaction (FCI) references without ever materializing dense $2^n \times 2^n$ matrices.

1 Executive Summary & Master Benchmark Results

The simulation pipeline executed the complete electronic structure sequence:

PySCF (RHF / STO-3G) $\longrightarrow$ FCIDUMP Records $\longrightarrow$ Jordan-Wigner CAR Mapping $\longrightarrow$ Sector FCI Reference

All four molecules were solved end-to-end within particle-number and spin symmetry sectors (N, S_z) without constructing dense $2^n \times 2^n$ matrices.

Table 1: Comprehensive Benchmark Results across Target Molecules

Molecule	Basis	Qubits	N_e	S_z	E_{RHF} (Ha)	E_0^{exact} (Ha)	Complete SD E_0 (Ha)	Error (mHa)	Chem. Acc.	SD Time	$\langle d \rangle$	$\langle S^2 \rangle$
LiH	STO-3G	12	4	0	-7.862023860	-7.882401932	-7.882388615	+0.0133	YES	269.8s (4.5 min)	0.333333	$< 10^{-7}$
BeH ₂	STO-3G	14	6	0	-15.560334936	-15.595182357	-15.594429802	+0.7526	YES	1881.9s (31.4 min)	0.428571	$< 10^{-37}$
HF	STO-3G	12	10	0	-98.570779986	-98.596624180	-98.596624180	+0.0000	YES (exact)	37.3s (0.6 min)	0.833119	$< 10^{-31}$
H ₂ O	STO-3G	14	10	0	-74.963023138	-75.012578241	-75.011873169	+0.7051	YES	1514.3s (25.2 min)	0.714286	$< 10^{-9}$

Key Finding: Every single target molecule achieves chemical accuracy (≤ 1.5936 mHa ≈ 1.0 kcal/mol). Hydrogen Fluoride (HF) matches exact sector FCI to machine precision (10^{-11} Ha). Lithium Hydride (LiH) achieves $120\times$ better than chemical accuracy (+0.0133 mHa). Beryllium Hydride (BeH₂) and Water (H₂O) achieve $\sim 2.1\times$ and $2.3\times$ better than the threshold, respectively.

2 Theoretical Architecture & Methodology

The `clifford_qc` simulation engine relies on five theoretical and engineering pillars:

1. **Sparse Multivector Engine (MV):** Quantum states (ρ), unitaries (U), observables (O), and CAR operators ($c_j, c_j^\dagger$) share a sparse Pauli-word representation in $\text{Cl}(2n, \mathbb{C}) \cong \text{M}(2^n, \mathbb{C})$.
2. **Binary-Symplectic Word Encoding:** Qubit Pauli letters are packed into 2 bits per qubit ($0 = I, 1 = X, 2 = Y, 3 = Z$), with multiplication computed via bitwise XOR, AND, and popcount (mod 4).
3. **Strict FCIDUMP Ingestion:** Restores packed chemist-notation integrals ($pq|rs$) into spin-orbital Jordan-Wigner multivectors without external third-party matrix dependencies.
4. **Sector-Restricted FCI Reference:** `SectorStatevectorBackend` stores statevector amplitudes strictly over $C(n, k)$ particle/spin symmetry sectors, establishing exact benchmark ground energies E_0^{exact} without dense 2^n -dimensional memory.
5. **Adaptive A-CASE & Complete SD Subspace Eigensolver:** Diagonalizes the Hamiltonian in an adaptively selected operator subspace basis $\{A_i|\psi_0\rangle\}$. Projected matrix elements $H_{ij} = \langle\psi_0|A_i^\dagger H A_j|\psi_0\rangle$ and overlaps $S_{ij} = \langle\psi_0|A_i^\dagger A_j|\psi_0\rangle$ are evaluated as single expectation values on reference state $|\psi_0\rangle$ without ever preparing basis states $A_i|\psi_0\rangle$ on hardware.

3 System Configurations & Geometries

- **Lithium Hydride (LiH):** Interatomic distance $d(\text{Li-H}) = 1.595 \text{ \AA}$. 12 qubits (6 spatial orbitals), Sector ($N = 4, S_z = 0$), Sector Dimension $\binom{6}{2}^2 = \mathbf{225}$. Total SD candidates: 92.
- **Beryllium Hydride (BeH₂):** Linear geometry $d(\text{Be-H}) = 1.326 \text{ \AA}$. 14 qubits (7 spatial orbitals), Sector ($N = 6, S_z = 0$), Sector Dimension $\binom{7}{3}^2 = \mathbf{1,225}$. Total SD candidates: 204.
- **Hydrogen Fluoride (HF):** Interatomic distance $d(\text{H-F}) = 0.917 \text{ \AA}$. 12 qubits (6 spatial orbitals), Sector ($N = 10, S_z = 0$), Sector Dimension $\binom{6}{5}^2 = \mathbf{36}$. Total SD candidates: 35.
- **Water (H₂O):** Bond distance $d(\text{O-H}) = 0.9575 \text{ \AA}$, angle $\theta = 104.51^\circ$. 14 qubits (7 spatial orbitals), Sector ($N = 10, S_z = 0$), Sector Dimension $\binom{7}{5}^2 = \mathbf{441}$. Total SD candidates: 140.

4 Detailed Per-Molecule Benchmark Results

4.1 Lithium Hydride (LiH)

- Hartree-Fock Energy: -7.862023860 Ha
- Exact Sector FCI Energy: -7.882401932 Ha
- Adaptive A-CASE Energy ($M = 11$): -7.862131491 Ha (Error: $+20.2704 \text{ mHa}$)
- **Complete SD Subspace Energy ($M = 93$): -7.882388615 Ha (Error: $+0.0133 \text{ mHa}$)**
- Dominant Ground-State Ritz Vector: $c_{\text{HF}} = +0.9999898$, $c_{E(4,5\leftarrow 0,1)} = +0.0040802$, $c_{E(6,7\leftarrow 0,1)} = +0.0017758$.
- Observables: $\langle d \rangle = 0.333333$, $\langle S^2 \rangle < 10^{-7}$, $\chi_0 = 4.25 \times 10^{-8}$.

4.2 Beryllium Hydride (BeH₂)

- Hartree-Fock Energy: -15.560334936 Ha
- Exact Sector FCI Energy: -15.595182357 Ha
- Adaptive A-CASE Energy ($M = 4$): -15.560359951 Ha (Error: $+34.8224$ mHa)
- **Complete SD Subspace Energy ($M = 205$): -15.594429802 Ha (Error: $+0.7526$ mHa)**
- Dominant Ground-State Ritz Vector: $c_{\text{HF}} = +0.9999987$, $c_{E(6,7 \leftarrow 0,1)} = +0.0015865$.
- Observables: $\langle d \rangle = 0.428571$, $\langle S^2 \rangle < 10^{-37}$, $\chi_0 = 7.22 \times 10^{-12}$.

4.3 Hydrogen Fluoride (HF)

- Hartree-Fock Energy: -98.570779986 Ha
- Exact Sector FCI Energy: -98.596624180 Ha
- Adaptive A-CASE Energy ($M = 15$): -98.576853731 Ha (Error: $+19.7704$ mHa)
- **Complete SD Subspace Energy ($M = 36$): -98.596624180 Ha (Error: $+0.0000$ mHa, exact sector match)**
- Dominant Ground-State Ritz Vector: $c_{\text{HF}} = +0.9988456$, $c_{E(10,11 \leftarrow 2,3)} = +0.0319772$.
- Observables: $\langle d \rangle = 0.833119$, $\langle S^2 \rangle < 10^{-31}$, $\chi_0 = 1.41 \times 10^{-8}$.

4.4 Water (H₂O)

- Hartree-Fock Energy: -74.963023138 Ha
- Exact Sector FCI Energy: -75.012578241 Ha
- Adaptive A-CASE Energy ($M = 7$): -74.963055822 Ha (Error: $+49.5224$ mHa)
- **Complete SD Subspace Energy ($M = 141$): -75.011873169 Ha (Error: $+0.7051$ mHa)**
- Dominant Ground-State Ritz Vector: $c_{\text{HF}} = +0.9999996$, $c_{E(10,11 \leftarrow 0,1)} = +0.0007123$.
- Observables: $\langle d \rangle = 0.714286$, $\langle S^2 \rangle < 10^{-9}$, $\chi_0 = 4.08 \times 10^{-12}$.

5 Resource Accounting & Performance

- **Pauli Word Universes:** LiH (42,460 terms), BeH₂ (21,930 terms), HF (75,950 terms), H₂O (97,498 terms).
- **Total Wall-Clock Runtime:** ~ 59 seconds for adaptive A-CASE; 3703.3 seconds (~ 61.7 min) for complete SD subspace eigensolves across all 4 molecules.
- **Artifact Inventory (molecular_results/):** FCIDUMP files (lih.fcidump, beh2.fcidump, hf.fcidump, h2o.fcidump), per-molecule JSON records, and master summary results_summary.json.

6 Conclusion

The end-to-end simulation demonstrates that `clifford.qc`'s A-CASE operator subspace solver and sector-restricted backends successfully ingest electronic structure FCIDUMP records, producing exact sector ground references, compact Rayleigh-Ritz eigensolutions, correlation observables, and dynamic Lehmann response spectra without materializing dense $2^n \times 2^n$ statevector matrices. **Chemical accuracy (≤ 1.5936 mHa) is achieved for all four benchmark molecules** when the subspace basis is expanded to the complete set of single-and-double excitations, with errors ranging from machine precision (HF) to +0.7526 mHa (BeH₂).